

Project Report

Introduction:

In an era marked by increasing digitalization and interconnectedness, safeguarding personal information has become more crucial than ever. One of the primary lines of defense against unauthorized access is the strength of user passwords. This project seeks to delve into the intricacies of password security by exploring various criteria such as length, complexity, and common patterns. The overarching question guiding this research is: What factors contribute most significantly to the strength of a password? To answer this question, we will employ a multifaceted approach, drawing insights from a dataset comprising randomly generated passwords and a list of the most common passwords identified in 2023.

Dataset(s):

The dataset used in this project serves as a microcosm of the diverse landscape of passwords employed by users. It consists of two distinct components:

000webhost Data Breach 2015:

When choosing a data set to analyze for this project, we were initially drawn to a list of the 10,000 most common passwords of 2023. After analyzing that data set, however, we discovered that most of those passwords are not a good random sample of passwords to use. This is because most of the common passwords are common words or phrases, and even the longer passwords in the dataset lack any complexity at all.

After a bit of research, we decided on a data set with a wider range of lengths and complexity. In October of 2015, hackers exploited a vulnerability within an old version of PHP used by the website hosting service 000webhost.com. 000webhost did not have their passwords properly secured, resulting in over 13 million plaintext passwords being leaked. The data set we used comes from a github repository that archives data from breaches like this for analysis purposes. Our data set contains a sample of just over 700,000 of these 13 million passwords. The passwords in the data set are of varying lengths and complexities, making it an ideal data set to analyze for our project.

By combining these two datasets, we aim to offer a comprehensive view of password security, encompassing both theoretically robust passwords and those that are commonly used but inherently weaker. This approach enables a nuanced analysis

of the factors that contribute to the strength or weakness of passwords, laying the groundwork for informed recommendations on password creation and management.

Randomly Generated Passwords:

This subset is generated using a combination of uppercase and lowercase letters, numbers, and symbols. The passwords vary in length, ranging from relatively short to longer, more complex strings. The randomness of these passwords ensures a representative sample of potential user-generated combinations.

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Methods:

Password Strength Checker: The Password Strength Checker tool employs a meticulous approach to assess the security of passwords. Here is a detailed breakdown of the methods used:

String Analysis: The password is dissected into a character array, allowing for a granular examination of each element. This step facilitates the subsequent analysis of unique characters and their frequency within the password.

Measurement of Unique Characters: By scrutinizing the character array, the tool determines the number of unique characters present in the password. A higher count of unique characters generally contributes to increased entropy, enhancing the password's resilience against common attacks.

Length Measurement: Simultaneously, the tool measures the length of the password, recognizing the fundamental role that length plays in security. Longer passwords inherently offer a larger search space for potential attackers, making them more robust against brute-force attempts.

Character Types and Patterns: Beyond uniqueness and length, the tool evaluates the presence of diverse character types, such as uppercase and lowercase letters, numbers, and special characters. Common patterns, like sequential characters or repeated sequences, are also identified, as these may weaken the overall security of the password.

Scoring System: Combining the insights from unique characters, length, and character types, the tool employs a scoring system to quantify the overall strength of the password. This scoring system aids users in understanding the relative security of their chosen passwords.

Comparison of Passwords: To delve into the trade-offs between longer, less complex passwords and shorter, more complicated ones, a detailed analysis was conducted:

Longer, Less Complex Passwords: A subset of passwords with increased length but reduced complexity was selected. The focus here was to explore the impact of a larger search space on security, even with simpler character compositions.

Shorter, More Complex Passwords: Conversely, a set of passwords with shorter lengths but heightened complexity was chosen. This comparison aimed to elucidate whether increased complexity could compensate for the reduced length in terms of security.

Time-to-Crack Assessment: Utilizing the time-to-crack calculation method, the estimated time required to crack passwords from both subsets was determined. This analysis sheds light on whether longer, less complex passwords or shorter, more complex ones are inherently "safer" from a security standpoint.

Graphical Representation: The results of this comparison are graphically presented, illustrating the nuanced relationship between password length, complexity, and overall security. This visual representation provides users with a clear understanding of the implications of their password choices.

Results

Through rigorous analysis utilizing our Password Strength Checker and a comprehensive study of diverse password types, our investigation unveiled profound insights into the intricacies of password security.

Password Strength Checker:

The intricate dissection of passwords into character arrays constituted the foundation of our exploration. This approach allowed for a meticulous examination of the interplay between uniqueness and frequency within passwords. Our scrutiny of unique characters within these arrays revealed a compelling correlation—an increase in the count of

unique characters directly correlated with heightened entropy. This correlation accentuated the significance of incorporating diverse character compositions in password creation. Simultaneously, our examination of password length illuminated its pivotal role in augmenting security. Longer passwords inherently offer a larger search space for potential attackers, thereby enhancing resilience against brute-force attempts. Additionally, our evaluation of character types and patterns within passwords unveiled vulnerabilities associated with commonplace sequences and repetitive structures.

This comprehensive analysis led to the development of a robust scoring system. This scoring mechanism serves as an intuitive metric, providing users with a tangible evaluation of their password's overall strength. The amalgamation of these factors enables users to garner actionable insights, facilitating the optimization of their password security strategies.

Comparison of Passwords:

Our in-depth exploration into the dynamics between longer, less complex passwords and shorter, more complex ones shed light on a pivotal aspect of password security. Despite their simplicity, longer passwords demonstrated a level of resilience owing to their expanded search space. However, our comparative analysis accentuated the irreplaceable role of complexity in fortifying passwords against sophisticated cyber threats. Shorter passwords with heightened complexity consistently emerged as stalwarts of security, challenging the conventional belief that length alone guarantees robustness.

The rigorous time-to-crack assessment further underscored the intricate balance between password length and complexity. While length significantly contributes to resilience, our findings unequivocally emphasize the indispensable nature of complexity in safeguarding passwords against modern cyber threats. This insight highlights the need for a multifaceted approach to password security—one that balances length and complexity to fortify against evolving security challenges."

Conclusion

In navigating the complex terrain of password security, our multifaceted analysis has yielded insightful observations that resonate across both theoretical and real-world contexts. The amalgamation of our Password Strength Checker and the comparative analysis of different password types provides a nuanced understanding of the factors influencing password security.

The Password Strength Checker, by breaking down passwords into character arrays, meticulously assessing unique characters, and measuring length, serves as a robust tool for users to gauge the resilience of their passwords. The incorporation of character types and pattern analysis adds an additional layer of sophistication, reflecting real-world attack vectors. The scoring system not only quantifies the strength of passwords but also empowers users with actionable insights for password improvement.

Our exploration of longer, less complex passwords versus shorter, more complicated ones delves into a pivotal conundrum faced by users – striking the delicate balance between memorability and security. The time-to-crack assessment unveils a compelling narrative: the intricate dance between password length and complexity. Longer, less complex passwords, with their expanded search space, showcase a degree of resilience, but our findings underscore the indispensable role of complexity in fortifying passwords. Shorter, more complex passwords consistently emerge as stalwarts of security, dispelling the notion that length alone is a panacea.

The graphical representation of our results paints a vivid picture of this intricate interplay. It not only serves as an educational tool for users but also reinforces the significance of adopting robust password practices. The visual depiction elucidates the logarithmic increase in security achieved by embracing complexity, offering a tangible benchmark for users seeking to fortify their digital fortresses.

In the broader context of cybersecurity, our project advocates for a paradigm shift in password creation and management. Common passwords, often birthed from user convenience, stand as Achilles' heels in the face of evolving threats. The project champions the adoption of randomly generated passwords and passphrase strategies, presenting them not as esoteric concepts but as pragmatic solutions accessible to all.

As we conclude this exploration, it becomes evident that enhancing password security requires a holistic approach. Educating users about the nuanced dynamics of password strength, providing tools for self-assessment, and promoting the adoption of resilient password creation strategies collectively form a formidable defense against unauthorized access. Our project, grounded in empirical analysis and user-centric principles, contributes to the ongoing dialogue on fortifying the digital realm, one password at a time.